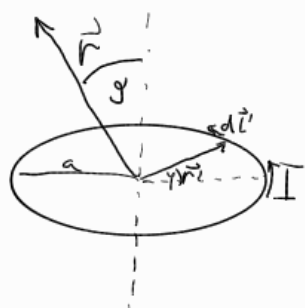


1) Magn. polje krožne tokovne zanke



$$\vec{A}(\vec{r}) = ?$$

$$r \gg a$$

$$\nabla \cdot \vec{B} = 0 \rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

$$\rightarrow \nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{j}$$

$= 0$ (izbira potenciala)

$$\Rightarrow \nabla^2 \vec{A} = -\mu_0 \vec{j}$$

↓ analogno z elektrost.

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{j}(\vec{r}') d^3 r'}{|\vec{r} - \vec{r}'|}$$

→ za vodniške (ni zvezno porazd. toka):

$$\vec{j} d^3 r' = \vec{j} s dl = I d\vec{l}$$

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l}}{|\vec{r} - \vec{r}'|}$$

$$\vec{r}' = a \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} \quad d\vec{l}' = a d\varphi \begin{bmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{bmatrix} \quad \vec{r} = r \begin{bmatrix} \sin y \\ 0 \\ \cos y \end{bmatrix} \leftarrow \text{postavimo } \vec{r} \text{ nad } x\text{-os zaradi osne simetrije}$$

$$|\vec{r} - \vec{r}'| = \left| \begin{bmatrix} r \sin y - a \cos \varphi' \\ -a \sin \varphi' \\ r \cos y \end{bmatrix} \right| = \sqrt{r^2 \sin^2 y - 2 r a \sin y \cos \varphi' + a^2 \cos^2 \varphi' + a^2 \sin^2 \varphi' + r^2 \cos^2 y} =$$

$$= \sqrt{r^2 + a^2 - 2 r a \sin y \cos \varphi'}$$

$$\frac{1}{|\vec{r} - \vec{r}'|} = (r^2 - 2 r a \sin y \cos \varphi')^{-1/2} = \frac{1}{r} \left(1 - 2 \frac{a}{r} \sin y \cos \varphi' \right)^{-1/2} \approx \frac{1}{r} \left(1 + \frac{a}{r} \sin y \cos \varphi' \right)$$

$$\rightarrow \vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi r} \int_0^{2\pi} a d\varphi \begin{bmatrix} -\sin \varphi' \\ \cos \varphi' \\ 0 \end{bmatrix} \left(1 + \frac{a}{r} \sin y \cos \varphi' \right) =$$

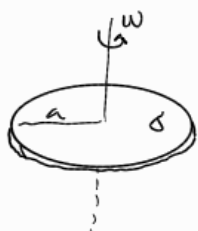
$$= \frac{a \mu_0 I}{4\pi r} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{1}{2} \frac{a}{r} \sin y \int_0^{2\pi} \cos^2 \varphi' d\varphi' =$$

$$= \frac{\mu_0 I}{4} \left(\frac{a}{r} \right)^2 \sin y \hat{e}_y = \frac{\mu_0}{4\pi} \underbrace{\frac{p_m}{r^2} \sin y \hat{e}_y}_{\hat{e}_z \times \frac{\vec{r}}{r}} = \frac{\mu_0}{4\pi} \frac{\vec{p}_m \times \vec{r}}{r^3} \quad \text{potencial mag. dipola}$$

$$\int_0^{2\pi} \cos^2 \varphi d\varphi = \frac{1}{2} \cdot 2\pi$$

$$\int_0^{2\pi} \sin \varphi \cos \varphi d\varphi = \int_0^{2\pi} \frac{1}{2} \sin 2\varphi d\varphi = 0$$

② Magnetno polje nabite vrteće se plošče



a) $B_z(z) = ?$

b) $p_m = ?$

c) $z \gg a$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{j}(\vec{r}') \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} d^3r'$$

$$dI = \frac{dq}{dt} = \omega \frac{dq}{2\pi} = \frac{\sigma 2\pi r' dr'}{2\pi} \omega = \sigma \omega r' dr'$$

$$\vec{j} \cdot d^3r' = j ds' \cdot r' d\varphi' = dI \cdot \vec{t}' r' d\varphi' = \sigma \omega r'^2 dr' d\varphi' \vec{t}'$$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \sigma \omega \int r'^2 dr' d\varphi' \frac{\vec{t}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \quad \vec{t}' = \begin{bmatrix} -\sin\varphi' \\ \cos\varphi' \\ 0 \end{bmatrix} \quad \vec{r} = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \quad \vec{r}' = r' \begin{bmatrix} \cos\varphi' \\ \sin\varphi' \\ 0 \end{bmatrix}$$

$$\vec{t}' \times (\vec{r} - \vec{r}') = \begin{vmatrix} i & j & k \\ -\sin\varphi' & \cos\varphi' & 0 \\ r'\cos\varphi' & r'\sin\varphi' & z \end{vmatrix} = \begin{bmatrix} z\cos\varphi' \\ z\sin\varphi' \\ r'\sin^2\varphi' + r'\cos^2\varphi' \end{bmatrix} = \begin{bmatrix} z\cos\varphi' \\ z\sin\varphi' \\ r' \end{bmatrix}$$

$$|\vec{r} - \vec{r}'|^3 = |r'^2 + z^2|^{3/2}$$

$$\vec{B}(z) = \frac{\mu_0}{4\pi} \sigma \omega \int \frac{r'^2}{(r'^2 + z^2)^{3/2}} \begin{bmatrix} z\cos\varphi' \\ z\sin\varphi' \\ r' \end{bmatrix} dr' d\varphi'$$

$$B_z(z) = \frac{\mu_0}{4\pi} \sigma \omega 2\pi \int_0^a \frac{r'^3}{(r'^2 + z^2)^{3/2}} dr' =$$

$$r'^2 + z^2 = u \\ du = 2r' dr'$$

$$= \frac{\mu_0 \sigma \omega}{4} \int_{z^2}^{a^2+z^2} \frac{u - z^2}{u^{3/2}} du = \frac{\mu_0 \sigma \omega}{4} \int_{z^2}^{a^2+z^2} (u^{-1/2} - z^2 u^{-3/2}) du = \frac{\mu_0 \sigma \omega}{4} \left[2u^{1/2} + 2z^2 u^{-1/2} \right]_{z^2}^{a^2+z^2} =$$

$$\stackrel{z \gg a}{\downarrow} = \frac{\mu_0 \sigma \omega}{2} \left[\sqrt{a^2 + z^2} + \frac{z^2}{\sqrt{a^2 + z^2}} - z - z \right] = \frac{\mu_0 \sigma \omega}{2} \left[\frac{a^2 + 2z^2}{\sqrt{a^2 + z^2}} - 2z \right]$$

$$z \gg a: \quad B_z(z) = \frac{\mu_0 \sigma \omega}{2} \left[\frac{a^2 + 2z^2}{z \sqrt{1 + \frac{a^2}{z^2}}} - 2z \right] \stackrel{k = \frac{a}{z}}{\downarrow} = \frac{\mu_0 \sigma \omega}{2} z \left[(k^2 + 2) \left(1 - \frac{1}{2} k^2 + \frac{3}{8} k^4 \right) - 2 \right] = \frac{\mu_0 \sigma \omega}{2} z \left[k^2 - \frac{1}{2} k^4 + \frac{3}{8} k^6 - k^2 + \frac{3}{4} k^4 - 2 \right] = \frac{\mu_0 \sigma \omega}{8} \frac{a^4}{z^3}$$

$$\vec{B}_{dipol} = \frac{\mu_0}{4\pi} \frac{3(\vec{p}_m \cdot \vec{r})\vec{r} - \vec{p}_m r^2}{r^5}$$

$$\vec{p}_m \parallel \vec{r}:$$

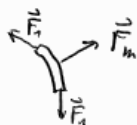
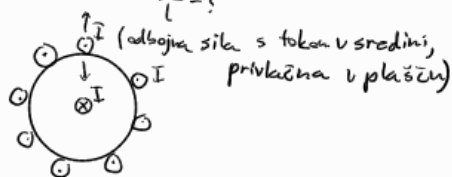
$$B(z) = \frac{\mu_0}{2\pi} \frac{p_m}{z^3}$$

$$\Rightarrow \frac{\mu_0 p_m}{2\pi} = \frac{\mu_0 \sigma \omega a^4}{8} \Rightarrow \boxed{p_m = \frac{\pi}{4} \sigma \omega a^4}$$

③ Mag. sila v koaksialnem kablu

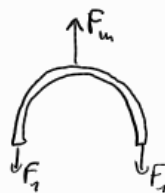
Sila napetosti plašča na dolžinsko enoto kabla = ?

$$\frac{F_t}{L} = ?$$



$$F_1 = \frac{1}{2} F_m$$

na pol plašča

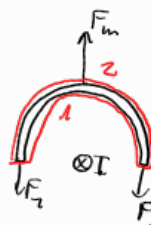


$$\vec{F}_m = \frac{1}{\mu_0} \oint [\vec{B}(\vec{B} \cdot \vec{n}) - \frac{1}{2} B^2 \vec{n}] dS$$

Ampere: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{tot}$

$$r > a: I_{tot} = 0 \Rightarrow B(r > a) = 0$$

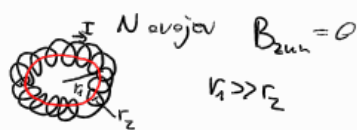
$$r < a: B 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$



$$\textcircled{1} B = 0 \Rightarrow F_m = 0$$

$$\begin{aligned} \textcircled{2} \vec{n} \perp \vec{B} \Rightarrow \vec{B} \cdot \vec{n} = 0 \Rightarrow \vec{F}_m &= \frac{1}{\mu_0} \int (-\frac{1}{2}) B^2 \vec{n} dS = -\frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi a} \right)^2 \int \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} a d\varphi dl = \\ &= \frac{\mu_0 I^2}{8\pi^2 a^2} (a) \int_0^\pi \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{bmatrix} d\varphi \\ F_{my} &= \frac{\mu_0 I^2}{4\pi^2 a} L \Rightarrow \boxed{\frac{F_t}{L} = \frac{\mu_0 I^2}{8\pi^2 a}} \end{aligned}$$

④ Mag. sila v toroidni tuljavi



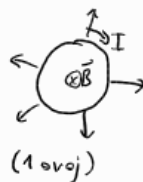
$$a) B_{not} = ?$$

$$b) F_1 = ? \text{ (sila napetosti ovojja)}$$

Ampere:

$$B \cdot 2\pi r = \mu_0 N I$$

$$\boxed{B = \frac{\mu_0 N I}{2\pi r_1}} \text{ (ker } r_2 < r_1)$$



$$\textcircled{2} B = 0 \Rightarrow \vec{F}_{m2} = 0$$

$$\textcircled{1} \textcircled{0}: \vec{n}_1 = -\vec{n}_2 \Rightarrow B \text{ isti} \Rightarrow \vec{F}_{m1} + \vec{F}_{m2} = 0$$

$$\textcircled{3}' F_{ms} = -\frac{1}{2\mu_0} \int B^2 \vec{n} dS =$$

$$= -\frac{1}{2\mu_0} \left(\frac{\mu_0 N I}{2\pi r_1} \right)^2 \hat{n}_s \cdot \vec{S}_{sp} = \left| S_{sp} \approx 2r_2 \frac{(2\pi r_1)}{N} \right|$$

$$= -\frac{\mu_0 N I^2 r_2}{2\pi r_1} \hat{e}_y$$